საქართველოს სამეფო სახლი

Royal House of Georgia

Digital Honor Authenticator

A System for Issuing and Verifying Royal Honors

This document explains how the Royal House of Georgia issues cryptographically verified honors and how anyone can confirm that a diploma is genuine.

March 2026

Fully operational as of 2026.

1. What Is This System?

The Royal House of Georgia issues honors (titles, orders, and distinctions) to individuals in recognition of distinguished service or merit. These honors are formally recorded on physical diplomas signed and sealed by the Head of the Royal House.

This system adds a layer of protection that a physical seal alone cannot provide: a mathematical proof of authenticity that anyone, anywhere in the world, can verify instantly by scanning a QR code on the diploma.

The proof is created by the Grand Master using a small hardware device called a YubiKey. The device never leaves his possession. Without it, no one can issue or forge a credential in the name of the Royal House.

In plain terms

Think of it as a wax seal that cannot be copied. Traditional seals can be replicated with the right tools and materials. This one cannot. The mathematics of the system make forgery computationally impossible, even for someone with access to powerful computers.

2. The Problem It Solves

The Royal House currently operates across multiple locations. The Head of the Royal House may pre-sign blank or partially-completed diplomas that are later filled in and issued by a representative in another country.

This creates a risk: a physical diploma bearing the signature of the Head of the Royal House could be issued fraudulently, or an authentic diploma could be altered after signing. A recipient, institution, or member of the public has no reliable way to confirm that a diploma is genuine.

The Risk	The Gap	The Solution
Anyone holding a pre-signed blank diploma could fill it in with any name and any honor.	There is currently no way for a recipient or institution to independently verify that the information on a diploma is what the Grand Master actually signed.	Each honor is digitally signed at the moment of issuance. The signature locks the exact recipient name, honor, specific distinction, and date. Any subsequent alteration invalidates it.

3. How It Works

The process has three steps, from issuance to verification.

Step 1 **Issue**	Step 2 **Print**	Step 3 **Verify**
The Grand Master fills in four fields: the recipient's name, the honor (selected from a list of recognized honors), the specific distinction or rank, and the date. He then touches his YubiKey to sign it. A QR code is generated as a vector file for the diploma designer to place and scale for print.	The QR code is printed on the physical diploma alongside the traditional seal and signature. The QR code carries the complete proof of authenticity. Nothing additional needs to be stored anywhere.	Anyone scans the QR code with a phone. A page opens immediately showing whether the diploma is authentic, and displaying the exact details that were signed.

Verification requires no app, no account, and no fee. The page is publicly accessible to anyone with an internet connection.

4. Verifying a Diploma

When someone scans the QR code on a diploma, their phone opens a page at:

https://verify.royalhouseofgeorgia.ge

The page performs a cryptographic check automatically and displays one of four possible results.

Authentic	Invalid	No Credential
The diploma is genuine. The page displays: • Recipient name • Honor title • Date issued • Signed by	The diploma failed verification. The page shows a clear rejection message. No credential details are displayed. This would occur if the diploma was altered, fabricated, or if the QR code was tampered with.	The page was opened directly, without a diploma QR code. This is what someone sees if they type the address into a browser rather than scanning a diploma. A network error or connection failure will also produce a distinct message on this page.

Revoked

The credential carries a valid signature but has been formally revoked by the Royal House. The page displays a clear revocation notice. This outcome is distinct from a forgery: the diploma was genuine when issued, but the Royal House has since withdrawn the honor.

Revocation is privacy-preserving: only an opaque hash of the credential is published, not the recipient's name or any other personal data.

Figure 1. Successful verification

Figure 2. Verification failure

Figure 3. Direct visit to page

5. The YubiKey: The Grand Master's Signing Device

The YubiKey is a small hardware device, similar in size to a USB drive, that stores the Grand Master's private signing key. It is the physical root of trust for the entire system.

Three properties make it the right tool for this purpose:

- The private key is generated inside the device and never leaves it. It cannot be copied, exported, or read by any software.
- Every signing operation requires the Grand Master's PIN and a physical touch of the device. An attacker who obtains the device cannot use it without the PIN. An attacker who knows the PIN cannot use it without the physical device.
- The device is small enough to carry at all times and costs less than $80.

What this means in practice

No one can issue a credential in the name of the Royal House without being physically present with the YubiKey and knowing the PIN. This is true regardless of how sophisticated the attacker is, or how many computers they control.

6. The Key Registry

The Grand Master's public verification key is the mathematical counterpart to the private key stored in the YubiKey. It is published openly in a registry at:

https://verify.royalhouseofgeorgia.ge/keys/registry.json

The public key does not enable anyone to sign credentials. It only enables verification. Publishing it openly means that anyone can confirm a signature is genuine without needing to contact the Royal House.

This is the core principle of public key cryptography. The system uses two mathematically linked keys: a private key that signs, and a public key that verifies. What one key locks, only the other can open. The private key never leaves the YubiKey. The public key can be given to the entire world without any loss of security, because it is computationally impossible to work backwards from the public key to reconstruct the private one.

Wide availability of the public key is therefore not a vulnerability: it is a design goal. The more widely it is distributed, the more independently verification can be performed, and the less the system depends on any single point of trust. A government ministry, a university admissions office, or a private individual can all verify a diploma without asking anyone for permission or assistance.

The registry is designed to be permanent. It records every key that has ever been used, with the dates during which it was active. This means:

- Credentials signed by the Grand Master today can still be verified in twenty years.
- If the Grand Master ever needs a new YubiKey, a new key is added to the registry with its start date. Old credentials remain valid under the old key.
- When signing authority passes to a successor, their key is added alongside the record of all previous authorities. Nothing is deleted.

7. Issuing an Honor: The Grand Master's Interface

The issuance interface is a self-contained desktop application installed on the Grand Master’s computer. It runs on macOS and Windows. It is not a website and has no server component. Before each signing, it connects to the internet to confirm that the YubiKey’s key is properly registered in the public registry. If the key is not registered, signing is refused.

The application has five sections:

- Sign: the credential form. The Grand Master fills in four fields: recipient name, honor (selected from a dropdown of recognized honors), specific distinction or rank, and date. He then touches the YubiKey to sign.
- History: a searchable record of every credential issued from this device, browsable by recipient name. A Revoke button allows the Grand Master to formally revoke any previously issued credential; this submits a pull request to the repository adding the credential’s hash to the revocation list.
- Registry: displays the public key registry. The Grand Master can import a new key directly from a YubiKey or from a certificate file, add or edit entries, and submit the updated registry for review. Changes are submitted as a GitHub pull request rather than committed directly; the repository administrator reviews and merges. GitHub login is handled via a one-time browser flow with no technical setup required, and the token is stored securely in the operating system keychain.

- Audit: shows the full commit history of the registry file on GitHub, providing tamper detection for the key record.
- YubiKey: confirms whether the currently inserted YubiKey is authorized in the registry, and displays its key details.

The application checks the registry over the internet on startup to confirm the YubiKey is authorized. Signing itself works offline: the YubiKey and the credential data are sufficient. If no internet connection is available, the registry check is skipped and a warning is shown, but the Grand Master can still issue credentials.

The Grand Master's complete signing interaction consists of three steps:

1. Fill in the four-field credential form: recipient name, honor, specific distinction or rank, and date.
2. **Click Sign Credential.**
3. Touch the YubiKey when prompted. The QR code appears immediately. It can be saved as a vector file (SVG) for the diploma designer, or as a PNG image for preview. The minimum recommended print size is 3 × 3 cm.

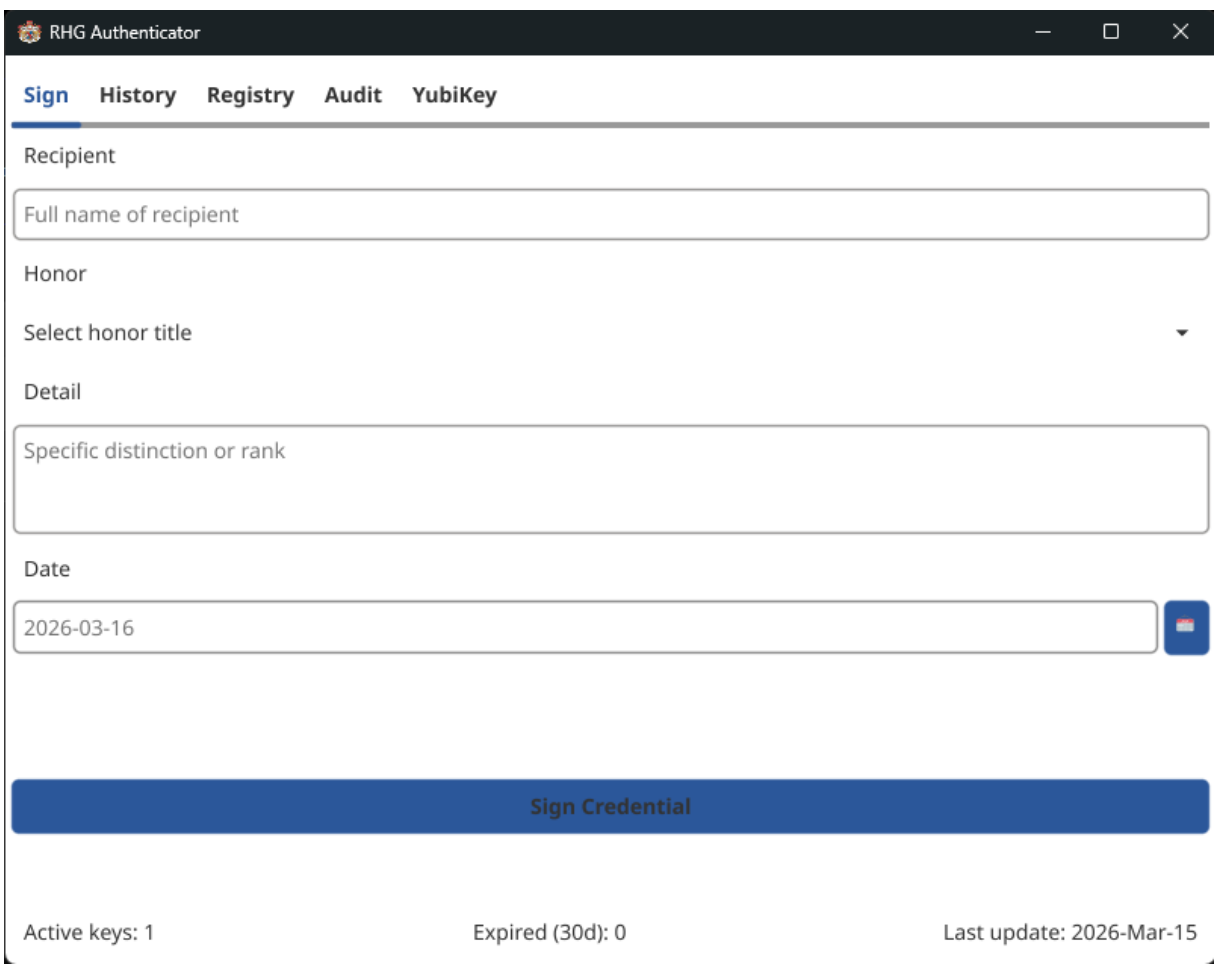

Figure 4. Issuance interface for the Grand Master

8. What Cannot Be Faked

The system was built with specific threats in mind. Here is how each one is handled.

Concern	How It Is Addressed
Someone issues a diploma without the knowledge of the Head of the Royal House	Impossible. Every credential requires the physical YubiKey and the PIN. No software workaround exists.
A genuine diploma is altered after signing	The signature covers every field of the credential: recipient name, honor, specific distinction, and date. Changing a single character invalidates the signature. Verification will fail.
Someone creates a fake QR code	The QR code encodes a mathematical proof. Without the private key, a valid proof cannot be constructed for fabricated data.
The verification website goes offline	The full cryptographic proof is embedded in the QR code URL. Technical users can verify independently using standard open-source tools, without the website.
The Grand Master loses his YubiKey	The old key is marked as retired in the registry. A new YubiKey is set up and its key added to the registry. Old credentials remain valid; new ones are signed with the new key.
Signing authority passes to a successor	The new authority's key is added to the registry with a start date. The historical record of previous keys is preserved permanently.
A credential needs to be revoked after issuance	The Grand Master uses the Revoke button in the History tab. This submits a pull request adding the credential's SHA-256 hash to the public revocation list. Once merged, verification returns a Revoked result. No personal data is published; only an opaque hash appears in the list.

9. What This System Does Not Require

Several technologies that might seem relevant to this kind of system were considered and ruled out. Each decision kept the system simpler and more durable.

- No blockchain. The cryptographic signature alone proves authenticity. A blockchain would add cost, complexity, and ongoing dependency without security benefit.
- No third-party verification service. There is no vendor that could shut down, change their pricing, or discontinue support.
- No app. Recipients and verifiers need only a standard phone camera.
- No account or registration. Anyone can verify any diploma at any time.
- No ongoing fees beyond the cost of maintaining the House's domain.

10. Privacy and Data

The system stores minimal data. The only information embedded in a credential is what is explicitly entered at the time of issuance: the recipient's name, the honor title, the date, and the signing authority.

No personal contact information, address, nationality, or any other data is collected or stored. The credential contains only what would appear on the face of the diploma itself.

The public verification page does not use cookies, collect analytics, or retain any record of who has verified a diploma or when.

11. Further Technical Detail

Full technical documentation is available in the project repository, including the cryptographic specification, module architecture, data formats, key registry schema, and developer setup guide.

The repository is publicly available. The system comprises a Go desktop application (the signing app) and a TypeScript verification library, with 1,034 automated tests between them (649 Go and 385 TypeScript).

Repository

https://github.com/RoyalHouseOfGeorgia/rhg_authenticator

Key documentation: ARCHITECTURE.md (system design and security model), DEVELOPER.md (TypeScript library reference), go/README.md (signing app build instructions and YubiKey setup). Release binaries are published with SHA-256 checksums and SLSA build provenance attestations for independent verification.

12. References

The following standards and specifications underpin the cryptographic and technical design of this system.

Edwards-Curve Digital Signature Algorithm (EdDSA)

RFC 8032. Internet Engineering Task Force (IETF), January 2017. rfc-editor.org/rfc/rfc8032

Defines the Ed25519 signature scheme used to sign and verify every credential issued by this system. Ed25519 was selected for its strong security properties, compact signature size, and resistance to implementation errors.

Personal Identity Verification (PIV) Standard

FIPS 201-3. National Institute of Standards and Technology (NIST), January 2022. Implemented in hardware by Yubico YubiKey 5 series (firmware 5.7+). doi.org/10.6028/NIST.FIPS.201-3

The PIV standard governs how cryptographic keys are stored and used on hardware security devices. The YubiKey implements this standard to ensure the Grand Master's private key is generated on-device, never exportable, and requires physical presence to operate.

Verifiable Credentials Data Model

W3C Recommendation. World Wide Web Consortium, March 2022. w3.org/TR/vc-data-model

An international standard for expressing cryptographically verifiable digital credentials. The credential format used by this system is informed by these principles, adapted for the specific operational requirements of the Royal House.

Public-Key Cryptography Standards

Diffie, W. and Hellman, M. "New Directions in Cryptography." IEEE Transactions on Information Theory, Vol. 22, No. 6, November 1976. doi.org/10.1109/TIT.1976.1055638

The foundational paper establishing public key cryptography. The principle described therein, that a key used to verify need not be kept secret, is the basis for the open publication of the Grand Master's verification key in the Royal House registry.